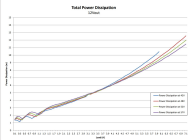
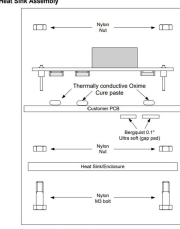
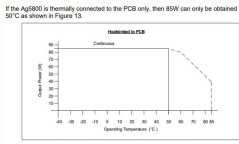


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AG5800 Thermal Considerations

AG5800 efficiency is 89% efficient. At 80W this would be 8.8W of heat loss

2.4.4 Output Power
The maximum available output power of the Ag5800 is limited by the available power supplied to the module. As a result, when powered by a DC supply the Ag5800 will be able to supply the rated 100W. However, when used with an IEEE802.3at PSE, which can only provide a maximum of 90W from its output, then the maximum achievable output power will be 85W.

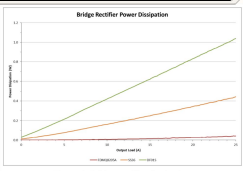


24V short J3 2 and 3

12V short J3 1 and 2 short J4 1 and 2

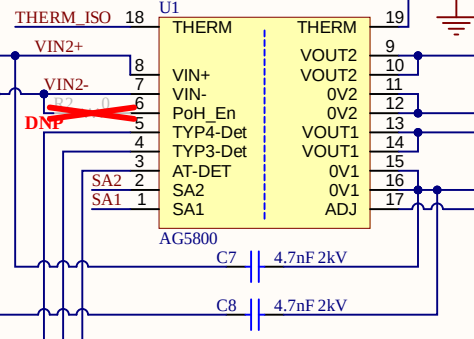


Each diode should see about 1A max which should be about 0.5V Vf. That is about 0.5W dissipation. Across 4 diodes = 2W loss

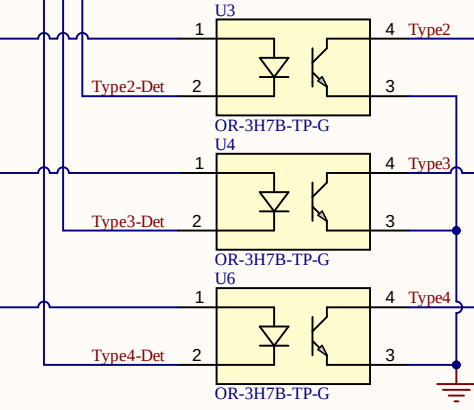


Vf = 700mV @ 3A

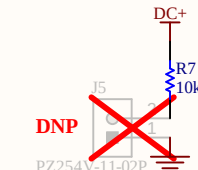
IEEE802.3bt 85W PD



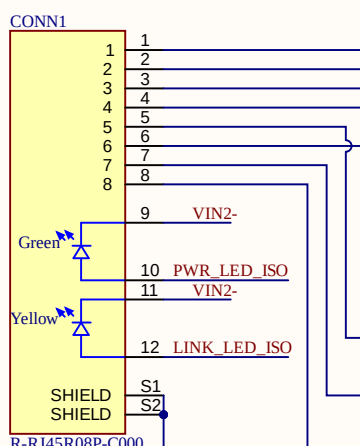
PoE-Type LED



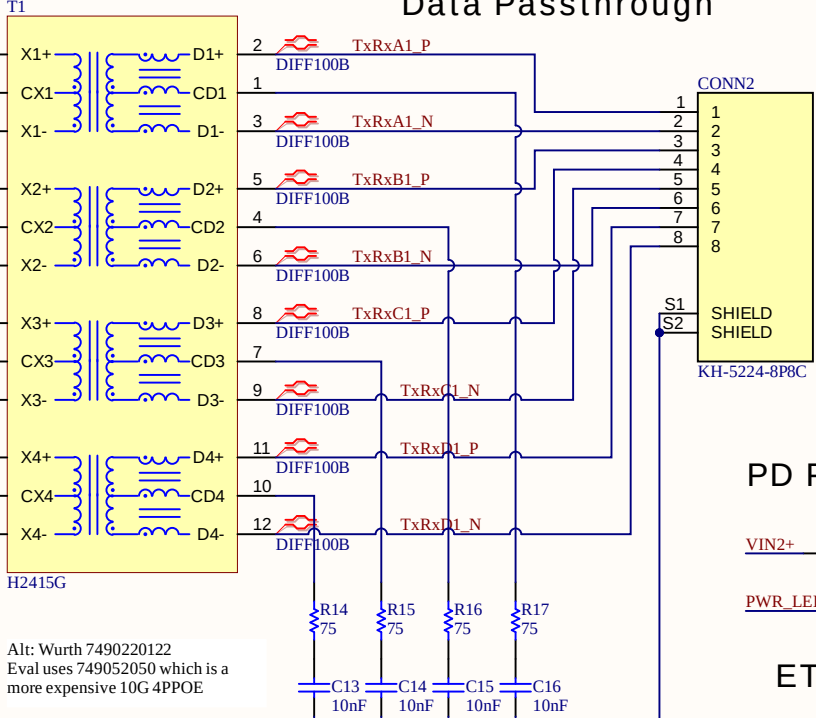
Optional DC+ LED



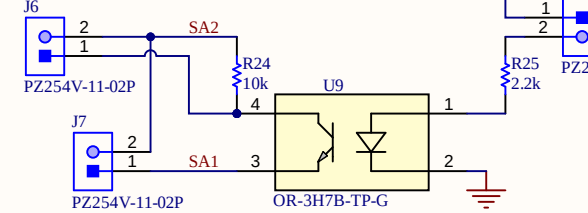
Power + Data Input



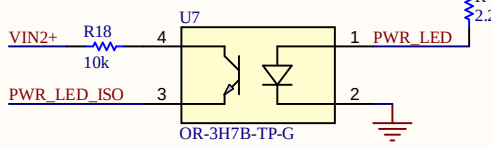
Data Passthrough



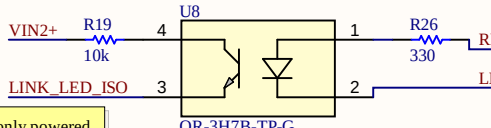
Auto PoE-PD Shut Off



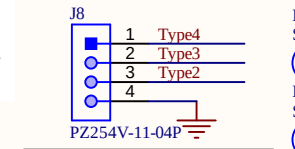
PD Power LED (RJ45 Green)



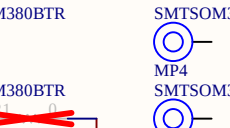
ETH Link LED (RJ45 Yellow)



External Pins



M3 Standoffs



BEOVOX-POE v2.1

Josh Horne

Basement Labs LLC

APPROVALS	DATE	PROJECT	Altium
ENG: *		PROJECT REVISION: *	
DSN: *		DOCUMENT REVISION: *	
CHK: *		DESIGN ITEM: *	
BOM:		TITLE	BEOVOX-POE
ASSY DWG:		SIZE	B
FAB DWG:		CAGE CODE	
PCB DWG:		DWG NO.	
		SCALE:	FILE NAME
			beovox-poe.SchDoc
		SHEET	1 OF 1